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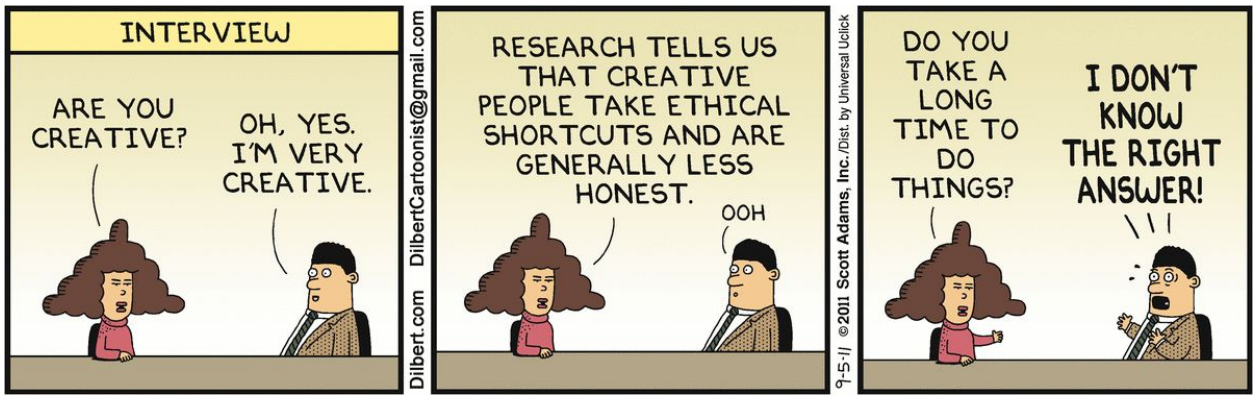
Data Analytics Exam

Do you have what it takes to be a Jampper?



Hi there, you have one day to complete the following exam. We will be evaluating the path to the proposed solution beside the result itself. So, as much as you can explain each step, the better. There are 3 exercises below, from coding to analytics.

Please use Python 3 in a Git repository. And remember that comments and cleansing means a lot. Hope you enjoy the process, *let's begin!*

1- Coding

Seems like the following function has an error. The function returns random values from a list, given the weights. For example: weighted_random([1, 2, 3], [0.5, 0.3, 0.2]), should return 1 with 50% of probabilities, 2 with 30% and 3 with 20%.

Objective: Evaluate if there's any error. If it has it, fix it with the proper explanation.

```
import random

def weighted_random(values, weights):
    total_weight = sum(weights)
    acum_weights = [w / total_weight for w in weights[:]]
    for i in range(len(weights)):
        acum_weights[i] += acum_weights[i]
    rand = random.random()
    for value, weight in zip(values, acum_weights):
        if weight > rand:
            return value
```

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2 - Analytics

Objective: Explain what the following SQL query does and how can be used as a trigger in the business side.
Plus: rewrite the query without the window function.

```
WITH events AS (
  SELECT
    device_id,
    time_stamp,
    LEAD(time_stamp) OVER (PARTITION BY device_id ORDER BY time_stamp) as next_time_stamp
  FROM events_table
  WHERE month = '201908'
  AND app_id = 1
  AND event_id = 4
),
per_event AS (
  SELECT
    device_id,
    DATE_DIFF('second', time_stamp, next_time_stamp) as time_diff
  FROM events
)
SELECT
  device_id,
  AVG(time_diff) as avg_per_user
FROM per_event
GROUP BY 1
```

Notes:

- Events: generic name for any action a user do in-app. Example: open, registration, add to cart, purchase.
- Device ID: is the unique identification for users.
- App ID: is the client Identification, is also unique per client
- Time Stamp: is the specific time an event happens.

3 - Math

"The Pirates law" says that if you have a treasure, the captain must choose a group of pirates and give the same amount of coins to each of them until there's no more coins available to give all of them exactly the same quantity of coins. So, the coins remaining will go to the captain's pocket.

Morgan (the captain) must split a treasure with less than 1000 golden coins. He knows that if he chose a group of 99 pirates, 51 coins are going to be for him and if he chose 77 pirates, only 29 coins will remain for him.

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Objective: How many pirates the Captain should select to achieve the maximum number of coins given “The pirate law”? And given the number of pirates, how many coins he will get?
Note: Pirates must have at least 1 coin each.

Please explain the logic process for the answer and justify it. Charts with explanations will be a nice Plus ;)